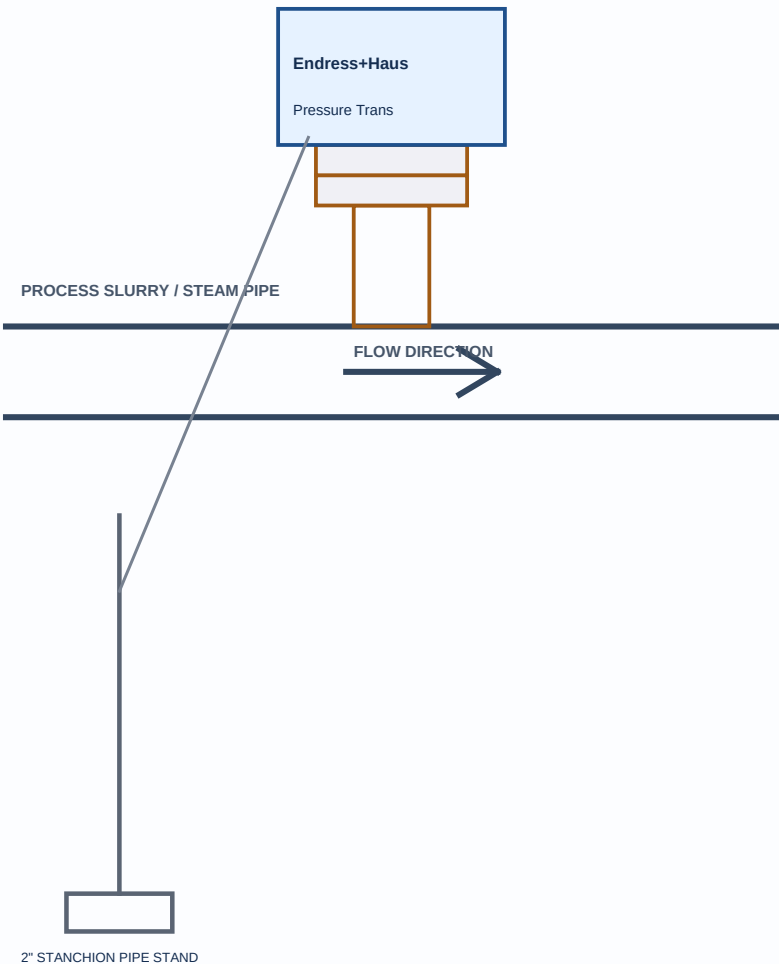


TEMPLATE TYP-17: PRESSURE TRANSMITTER

TYPICAL TAGS: PT-40101 / PT-60202

SERVICE: Continuous Pressure Monitoring for Cooker & Flash Tank | PLC I/O: 1756-IF16 (16-Point Differential AI)

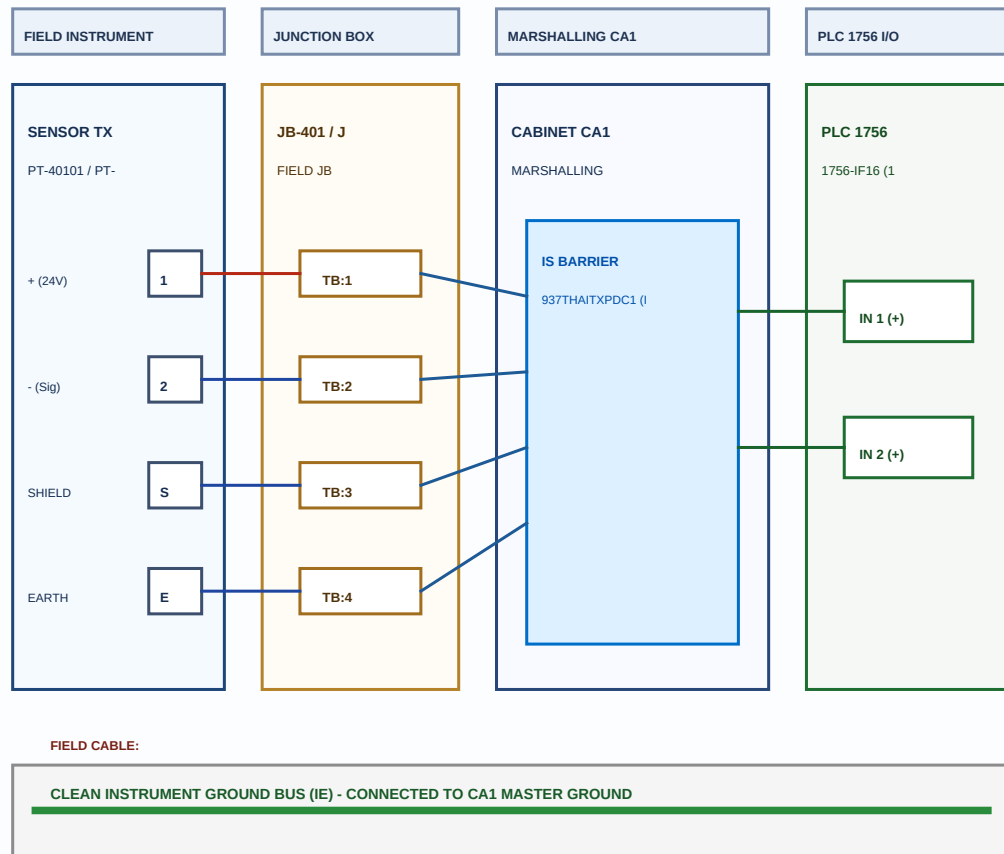
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Thread / 2" Tr..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART (0-10 bar / 0-16 bar gauge)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Shielded (Belden 8761)
Destination JB:	JB-401 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Differential AI)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Cerabar PMP51B / PMC51B (Metallic / Ceramic)

SERVICE: Continuous Pressure Monitoring for Cooker & Flash Tank

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Cerabar PMP51B Pressure Transm E+H Piezoresistive SS316L Cell		1 EA
2	2-Valve Block & Bleed Manifold SS316 1/2" NPT Male x Female		1 EA
3	2" Pipe Mounting Bracket Kit SS304 Universal Bracket & U-Bolt		1 SET
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-plated Brass Armored Gland		1 EA
5	Siphon / Cooling Neck (Steam D SS316 Pigtail Loop 1/2" NPT		1 EA

ENGINEERING INSTALLATION NOTES

- Use flush diaphragm seal (PMC51B) for thick starch slurry to eliminate clogging in impulse ports.
- Ground transmitter body to local pipe ground; shield wire grounded only at CA1 cabinet earth bar.
- HART configuration: Set damping to 1.0s to filter process pump pulsations.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONPRESSURE TRANSMITTER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-017
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
19 / 25